

UNIT 14: POLYMERS

Advanced Level Science for Technology (SFT) - Theory & Comprehensive Guide

AUTHOR: RAVINDRA GAMAGE | ENGLISH TRANSLATION & COMPILATION

Introduction to Polymers

Small molecules known as monomers are organic or occasionally inorganic structures consisting of small molecular frameworks formed when hydrogen atoms combine with carbon atoms possessing a double bond between two carbon atoms ($C=C$).

Monomer (Ethylene): $CH_2 = CH_2 \rightarrow$ Repeating Unit $\rightarrow$ Polymer: $[—CH_2—CH_2—]_n$

Repeating Units and Macromolecules (Polymers)

The double bond in the simple molecule uncoils to form the repeating unit. The joining of these repeating units yields polymer molecules. These can exist in linear, cross-linked, or cyclic structural configurations.

Types of Polymerization

- Addition Polymerization:** Numerous monomer molecules combine to form a single large polymer product without any byproduct. This is typically a long-chain product where all atoms present in the monomers are retained in the polymer. The process occurs in steps: chain initiation, chain propagation, and chain termination.
- Condensation Polymerization:** When a large number of monomer molecules combine to form a polymer molecule, if one or more byproducts (e.g., H_2O , HCl , etc.) are formed—meaning not all atoms from the monomers are found in the polymer—such polymers are called condensation polymers.

1. Polyethylene (Polythene)

The monomer molecule is ethene ($CH_2=CH_2$). Instead of the double bond between the two carbon atoms ($C=C$), a single bond remains, and the two electrons create two new bonds on either side.

- **High Density Polyethylene (HDPE):** Straight linear chain polythene produced at approximately $100^\circ C$ and low pressure (around 1 atmosphere).
- **Low Density Polyethylene (LDPE):** Branched-chain polythene produced at temperatures between $100 — 300^\circ C$ and high pressures between $1500 — 3000$ atmospheres.

General Properties of Polythene: Colorless (can be pigmented if needed), no specific odor, non-toxic, extremely slow decomposition rate (making it a major environmental pollutant), and low cost.

2. Polyvinyl Chloride (PVC)

Polyvinyl chloride is formed by the polymerization of numerous vinyl chloride molecules under high pressure at temperatures around $70 - 80^{\circ}\text{C}$.

- **Monomer:** Vinyl chloride ($1 \text{ ext{-chloroethylene}}$) or chloroethylene, commonly known in industry as Vinyl Chloride Monomer - VCM). Formed by replacing one H atom in ethene with a Cl atom ($\text{CH}_2=\text{CHCl}$).
- **Historical Context:** Accidentally discovered by French chemist Henri Victor Regnault in 1835 and German chemist Eugen Baumann in 1872. Commercial industrial use began around 1926.
- **Applications:** Water pipes and gutters, thin film sheets, electrical wire insulation, printing inks, sticker sheets, building materials (doors, windows), various footwear and apparel, toys, pens and ink tubes, single-use medical syringes, saline bottles, and waste containers.

3. Natural Rubber

Natural rubber originates from milky latex extracted from *Hevea brasiliensis*, a tree native to Brazil. Introduced and planted in Sri Lanka at Henarathgoda Botanical Garden, Gampaha.

- **Monomer:** Isoprene ($\text{ext{cis} ext{-isoprene}}$ or $2 \text{ ext{-methyl} ext{-1,3 ext{-butadiene}}}$).
- **Elasticity:** Due to cross-links (van der Waals forces) between rubber (polyisoprene) chains, its structure adopts a spiral conformation. Stretching occurs under applied force, returning close to its original form upon release.
- **Vulcanization:** Originally developed by Charles Goodyear. Rubber is reacted with small percentages of sulfur by weight to strengthen cross-links, preserving elasticity while reducing wear and increasing strength.
- **Applications:** Footwear, tires and tubes, blankets, raincoats, hoses, engine belt pulleys, balloons, cushions and mattresses, balls, liquid/chemical storage containers, gloves, adhesives, erasers, wrapping materials, phone accessories, computer/electrical components, vibration dampers, and washers.

4. Synthetic Rubber (Polychloroprene / Neoprene)

Manufactured synthetically using chloroprene ($2 \text{ ext{-chloro} ext{-1,3 ext{-butadiene}}}$).

- **Monomer:** Chloroprene ($\text{CH}_2=\text{CCl-CH=CH}_2$).
- **Applications:** Sound insulation materials, hoses, engine gaskets (packings), fire-resistant emergency exit doors (superior fire resistance due to Cl content), high-temperature metal surface sealants, low-cost fishing accessories, shock absorbers, footwear, adhesives, sporting goods, hydroponic water storage containers, and floor carpets.

5. Polystyrene

Discovered in 1839 by Eduard Simon. Typically a transparent glass-like material (e.g., CD cases). Familiar expanded polystyrene (Styrofoam) is produced by heating and liquefying with blowing agents to create gas bubbles before solidification.

- **Monomer:** Styrene (vinylbenzene, $C_6H_5CH=CH_2$).
- **Properties:** White solid, lightweight, excellent thermal and electrical insulator.
- **Applications (Standard):** CD/DVD/BD cases, fire smoke detector housings, automobile license plates, scale models (vehicles, ships, buildings), and razors.
- **Applications (Styrofoam / Expanded):** Thermal insulation for ice/ice cream, float components for fishing nets, cake and model bases, food packaging boxes, decorative items, and sound insulation.

6. Teflon (Polytetrafluoroethylene - PTFE)

- **Monomer:** Tetrafluoroethylene ($CF_2=CF_2$).
- **Applications:** Non-stick cookware and sealing tapes (hydrophobic), high-wear automotive/machinery components, corrosive chemical storage/valves/gaskets, high-temperature machinery/oven components, and fire-resistant safety clothing.

7. Condensation Polymers

Formed by the combination of numerous monomers with the elimination of small byproduct molecules (such as H_2O or HCl).

Urea-Formaldehyde

Produced by reacting urea ($NH_2-CO-NH_2$) and formaldehyde ($HCHO$) in the presence of concentrated H_2SO_4 . It forms a three-dimensional thermosetting network.

- **Applications:** Furniture manufacturing, chipboard adhesives, and electrical appliance components.

Phenol-Formaldehyde (Bakelite)

The world's first synthetic thermosetting polymer (1907). Produced by reacting phenol (C_6H_5OH) and formaldehyde in the presence of concentrated H_2SO_4 .

- **Applications:** Electrical switches and plugs, automotive components, circuit boards, buttons, and billiard balls.

Polyesters (Terylene / Dacron / PET)

Formed by condensation of dicarboxylic acids (e.g., terephthalic acid) and diols (e.g., ethylene glycol / *ethane-1,2-diol*), linked by ester bonds (—COO—).

- **Applications:** Synthetic fabrics and threads, safety helmets, protective covers, drinking water bottles, and soft drink bottles.
- *Note on Methyl Salicylate:* Formed from salicylic acid and methanol; used in pain relief balms (Wintergreen), chewing gum, insect attractant pheromones, and mouthwashes.

Polyamides (Nylon)

Most commonly represented by Nylon-6,6 (formed from hexamethylenediamine and adipic acid) and Nylon-6.

- **Applications:** Synthetic ropes, footwear, guitar strings, fishing gear, toothbrush bristles, tire cords, tents, and water-resistant swimwear.

Comparative Polymer Classifications

Natural Polymers	Synthetic Polymers
Used since ancient times.	Introduced in recent times.
Biodegradable; environmentally friendly.	Mostly non-biodegradable; causes pollution.
Naturally recycled, maintaining ecological balance.	Requires human intervention for recycling; disrupts balance.
Produced exclusively within living cells (e.g., rubber, proteins, starch, cellulose).	Created in abiotic environments (e.g., polythene, synthetic rubber, plastics).

Addition Polymers	Condensation Polymers
Formed by addition of identical monomers.	Mostly formed by different reacting monomers.
No byproducts formed.	One or more byproducts formed (e.g., H_2O).
All monomer atoms retained in the polymer.	Polymer atom count is less than total monomer atoms.
Formed via monomer double bonds (e.g., rubber, PVC, polythene, teflon).	Formed via functional group reactions (e.g., nylon, bakelite, polyester).

Thermosetting Polymers	Thermoplastic Polymers
High melting and boiling points.	Lower melting and boiling points.
Do not melt easily upon heating.	Melt easily upon heating and can be reshaped.
Cross-linked structures present.	No cross-links (linear/branched).
Three-dimensional polymeric networks.	Mostly linear polymers.
Difficult to recycle; expensive.	Easy to recycle; economical (e.g., nylon, polystyrene, teflon, polythene, PVC).

Environmental and Human Impact

1. Non-biodegradable polymers accumulate in the environment, causing severe pollution.
2. PVC, polycarbonate, and polystyrene can be harmful to human and animal health, potentially causing illness or mortality.
3. Phthalates added as plasticizers act as endocrine disruptors, altering hormonal functions in the body.
4. Kidney filtration of accumulated toxins can lead to renal disorders and carcinogenesis.
5. Open burning of plastics releases hazardous, far-traveling carcinogenic gases.
6. Polythene exhibits relatively lower toxic emission risks compared to certain other synthetic polymers.

Mitigation Strategies

- Increase utilization of biodegradable materials.
- Promote multi-use and reuse of plastic products.
- Maximize recycling practices and establish mandatory recycling regulations.
- Enforce legislative restrictions on manufacturing and usage.
- Raise environmental awareness starting at the school level to foster cultural and attitudinal change.

Self-Assessment Review Question 01 (Structured)

A. Write balanced chemical equations for forming a thermoplastic polymer via (i) Addition Polymerization and (ii) Condensation Polymerization using the provided chemical species: $\text{CF}_2=\text{CF}_2$, $\text{CH}_2=\text{CH}-\text{CH}_3$, $\text{HO}-\text{CH}_2-\text{CH}_2-\text{OH}$, $\text{C}_6\text{H}_5\text{OH}$, $\text{H}_2\text{N}-(\text{CH}_2)_6-\text{NH}_2$, $\text{HOOC}-(\text{CH}_2)_4-\text{COOH}$. Clearly show repeating units and reaction types.

B. For Polystyrene, PVC, and Natural Rubber: (i) Write repeating unit structures, (ii) Identify monomer names, and (iii) Describe a simple home test to distinguish PVC from Bakelite.